

Khin Josmar V. Misuari

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RESEARCH INTERESTS

General Relativity · Early Universe Cosmology · Quantum Gravity Phenomenology · Gravitational Waves

EDUCATION

Abdus Salam International Centre for Theoretical Physics Trieste, Italy

Incoming Postgraduate Diploma in High Energy, Cosmology, and Astroparticle Physics (HECAP) Sept. 2026 –

- One of 10 students selected globally for the HECAP field; 5th Filipino admitted to HECAP since the program's inception and the first from the University of San Carlos and Cebu province.

University of San Carlos Cebu City, Cebu, Philippines

Bachelor of Science in Applied Physics Aug. 2019 – Jun. 2023

- **Cumulative GPA: 1.42 / 1.00** (1.00 = Highest/Excellent; 3.00 = Minimum Pass)
- **Graduate-Level Coursework** (non-credit, audited):
 - **General Relativity** · Grade: **1.00 / 1.00**, Ranked **1st** among M.Sc. & PhD candidates · SY 2021–2022, 2nd Semester
 - **Mathematical Methods in Physics** · Grade: **1.00 / 1.00**, Ranked **1st** among M.Sc. & PhD candidates · SY 2021–2022, Summer Term
- **Theoretical Electives:**
 - **Classical Mechanics II** (PHYSEL 4105) · Grade: **1.00 / 1.00**
 - **Quantum Mechanics II** (PHYSEL 4206) · Grade: **1.00 / 1.00**

Carcar City Alternative Learning System Carcar City, Cebu, Philippines

High School Diploma via Alternative Learning System (ALS) Mar. 2019 – Jun. 2019

- Bypassed four years of traditional high school through the national Accreditation & Equivalency (A&E) Test, advancing directly from Grade 8 to university.

RESEARCH EXPERIENCE

OJT Research Project Jun. 2022 – Jul. 2022

University of San Carlos · Supervisor: Dr. Roland Emerito S. Otadoy Cebu City, Philippines

- Independently reproduced results of a peer-reviewed publication in **quantum gravity phenomenology**: Abac, Esguerra, & Otadoy, *Int. J. Mod. Phys. D* **30** (2021) 2150005. Derived GUP-modified stellar structure equations and numerically reproduced mass-radius relations for zero-temperature white dwarfs.

Undergraduate Thesis · 2nd Place Best Thesis Award Aug. 2022 – Jun. 2023

University of San Carlos · Adviser: Dr. Danilo M. Yanga Cebu City, Philippines

- *Calculation of the Eliashberg Function of Particles in Graphene in the Migdal-Eliashberg Theory of Superconductivity*. Derived analytic expressions for the Eliashberg function using many-body formalism. Recognised with 2nd Place Best Thesis Award, Department of Physics.

ACADEMIC LEADERSHIP & PROGRAM DEVELOPMENT

Initiator & Program Head, ASCEND: Physics Jul. 2023 – May 2026

Sacred Heart School – Ateneo de Cebu Mandaue City, Cebu, Philippines

- Designed and led an original enrichment curriculum bridging high school to university-level physics and mathematics for students in Sacred Heart School – Ateneo de Cebu.
- Selected outcomes since program inception:
 - **Philippine Physics Olympiad (PPO) 2026: 4 out of 32 National Finalists — 2nd highest nationwide**, behind only Philippine Science High School Main Campus. **7 Qualifying Round Merit Awards — 1st in the Visayas, 3rd nationally** per school.
 - **PPO 2025: 2 out of 27 National Finalists**.
 - **National podium finishes:** UP Big Bang Astronomy Quiz Show 2026; UP PISIKaalaman 2026; and Ateneo Physics Olympiad 2024.

Member, Carolinian Physics & Astronomy Society

University of San Carlos

Aug. 2019 – Jun. 2023

Cebu City, Cebu, Philippines

- **Chairman of Academic Affairs, SY 2021–2022** · Led several structural departmental reforms such as inclusion of Classical Mechanics II as an elective in the B.Sc. Applied Physics program and structural revisions to the Mathematical Physics syllabus, re-sequencing advanced topics to better align with prerequisites for theoretical research and permanently expanding the theoretical coursework available to students.
- **Core Member, USS Quiz Bowl Series Question Committee** · Authored rigorous, university-level problems for the annual USC Physics and Astronomy Quiz Bowl inter-school competition.

PROFESSIONAL EXPERIENCE

Senior High School Faculty & Olympiad Head Coach

Sacred Heart School – Ateneo de Cebu

Jul. 2023 – May. 2026

Mandaue City, Cebu, Philippines

- **DOST-JLSS Return Service Obligation.** Fullfilled mandatory national scholarship service through high-level STEM instruction and talent development.
- **Head Coach, Physics & Astronomy Olympiad Team** · Pioneered academic competition culture at Sacred Heart School – Ateneo de Cebu.
- Instructor in General Physics 1 & 2 and Engineering Physics electives.

Private Tutor

Freelance

Jun. 2023 – Present

Cebu, Philippines

- Tutors subjects ranging from high school mathematics and physics to advanced upper-level topics.

Teaching Assistant (TA)

University of San Carlos, Department of Physics

Jan. 2023 – May. 2023

Cebu City, Philippines

- Selected by physics faculty to assist in Engineering Physics courses; assisted in delivering and grading assessments.

RELEVANT WORKSHOPS & TRAINING

1st Jagna Space Encounters | *Central Visayan Institute Foundation, Bohol*

Jan. 2025

- Intensive workshop on astrophysics and space science. Developed practical skills in Python for astronomical data analysis and computational modelling.

DISTINCTIONS & AWARDS

2020 – 2023 Dean's Honors List · **Ranked 10th (2020), 1st (2021), 2nd (2022)** · University of San Carlos

2023 2nd Place Best Thesis Award · Department of Physics, University of San Carlos

2022 Iron Wood Property Ventures Corporation Scholarship (merit-based)

2021 DOST Junior Level Science Scholarship (DOST-JLSS) · Granted to top STEM undergraduates in the Philippines

2019 ALS National Top Achiever · Exceptional score on A&E Test; advanced from Grade 8 directly to university

TECHNICAL SKILLS

Programming: Python, R, MATLAB

Data Analysis: pandas, NumPy, SciPy, dplyr, Excel

Data Visualisation: Matplotlib, ggplot2

Developer Tools: VS Code, PyCharm, Jupyter Notebook

Scientific Writing: $\text{\LaTeX}$, technical writing, experiment design

Languages: Cebuano (native), Filipino (fluent), English (fluent)